

CS4174 Performance Technology - Written Report

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ABSTRACT

Max MSP opens numerous opportunities in allowing its users to push audio and visual boundaries, as was the case in our own project which will be laid out in this report. Max MSP's strength of allowing these users to explore interactability in such a short period of time as opposed to other means of doing so has captured our interest as a group, and we sought to explore the capabilities of Max through augmenting an instrument's sound and volume through sensors that were provided through an Arduino kit.

1. INTRODUCTION

The following sections in this written report are laid out as follows: the idea and motivation behind the overall direction and artistic aim of our project, our process for which the project was planned and organised, the details of the system built in order to ensure the project's functionality and the subsequent reflection, as well as our reflection on said system. Furthermore, our plan for our performance as well as the reflection on said performance is laid out. Finally, we discuss how the skills we have developed through usage of Max MSP can be relevant to our respective career paths.

2. IDEA AND MOTIVATION

The initial idea for this project was developing something that could meld the worlds of music and technology into a usable tool for musicians. The motivation for this project was also twofold. First, following the theme of the combination of music and technology, our motivation was to make something a musician could feasibly use to play in a live performance, whilst adjusting the loop of an audio recording from a music instrument using natural body movements. These body movements would then affect the loop, visually melding the worlds of physical art and computing. The second motivation for this project was creating an augmented instrument. Putting aside the theme of bringing two worlds together, we also thought that creating an augmented instrument would be a useful experiment in creativity, and that it might give us a deeper understanding of the mechanics of creating an instrument, as well as a deeper understanding of music itself. We began to consider the options for an augmented instrument, by firstly doing some research. A big inspiration was the Augmented Instruments Laboratory (instrumentslab.org). They have many papers on the topic and we were able to gain a lot of information, like "One potential solution that might provide the

kind of nuanced analysis required for the study of expressive gestures is the use of complementary direct sensing methods that provide measures related to the physical features of the performer's actions" (Temprano, McPherson, 2021). This was in relation to what the author thought they could change about their project as a solution, and something we implemented in our own project by making the instrument respond to natural body movement.

3. PROCESS

The process for our project's development went through numerous iterations before having reached its final form, cultivating in our performance. We both agreed that the incorporation of Hugh's electric guitar, his Audio Go Box, as well as the button and accelerometer sensors attached to the guitar would be implemented. This concept then persisted into the final iteration as a result. The Arduino's button sensor would serve to control and loop the recording of Hugh's guitar playing. The sound would be transmitted into Max MSP via the Audio Go Box connected to the guitar and Tommy's laptop. The accelerometer sensor would control the recording's playback volume from the X-axis feedback (moving up and down). However, the accelerometer sensor's purpose was further expanded from the original iteration as laid out in our proposal to also include the ability to manipulate effects of the audio. To accomplish this, we used the Z-axis feedback and incorporated the Valhalla Supermassive plugin on Max MSP, which we felt would lend itself more to our artistic intent. There was a visual element that did not make it into the final iteration of this project with regards to the inclusion of visual effects in Max MSP. Initially, Tommy looked into the use of jitter objects on Max MSP with the idea of manipulating visuals on the screen in which the accelerometer-altered volume of the recorded loop would manipulate the visuals to respond to the volume accordingly. In the end, though, we both decided that we wanted our performance to be more artistically focused on *augmenting* the sound of instruments, as opposed to showing visuals on the screen that we would have felt were more redundant and not particularly relevant to our artistic intent. Furthermore, in an early iteration of our idea, we both wanted to initially include the accelerometer sensor on the neck of the guitar as opposed to where it would be placed towards the bottom front of the guitar neighboring the button as was demonstrated in our performance. We realized two issues with this plan: first, there would be no need to include the accelerometer on the neck of the guitar due to its sensor readings still being very well able to be manipulated

and detected even if it was simply placed on the bottom of the front of the guitar. Secondly, we felt that connecting a wire from the Arduino kit all the way up to the neck of the guitar would be far more complicated to reach, as well as more uncomfortable for Hugh to play. To determine the placement of the sensors, we sketched out the design and how they would appear on the guitar and found that there were flaws in our original plan. So, in the final iteration, the Arduino kit was placed on Hugh's lap while the wires were easily able to reach the sensors, which we found easily attachable with Blu Tack. This helped Hugh's ease of playing tremendously.

4. SYSTEM BUILT

Tommy first ensured that the required sensors were physically wired to the Arduino board and that the Arduino code was reading the relevant sensors and sending the required output to the serial port. Using the button's sensor input, a toggle object would alternate between "on" and "off" each time the button sensor was pressed. When turned on, the process to start recording begins and when turned off, the process to stop recording and immediately begin playback begins. The process to begin recording is initiated when the toggle outputs a '1' value. This value was sent to a trigger object that Tommy added that performs two bang operations from right to left. The first bang was sent to two messages: to a "stop" message, which is then sent to the groove object to stop anything that is currently being played back, and to a "clear" message, which was sent to the buffer to clear the contents of the buffer. Then, a second bang from the trigger was sent to a timer object, which was used to capture the start time of the recording. On the other hand, the process to end the recording and initiate the playback occurred when the toggle was turned off and a '0' was the output. A signal is sent to the right input port of the timer object that stops the timer and triggers the output from the timer. The output from the timer is the duration of the recording in milliseconds. A prepend object was then used by Tommy to build a message that is sent to the groove object, which specifies the length of time to play back the recording from the buffer containing Hugh's recording. Without the use of the timer object, the entire contents of the buffer would be played back, and we both agreed that this was undesirable if our planned recording for the performance was to be shorter than the time allowed in the buffer. Tommy included the groove object in order to play back the guitar's audio recording from the buffer. He used the 'attrui' object in order to set the "loop" attribute to keep the playback constantly looping. When the button was pressed and the recording stopped, the toggle output a '0' value that was used to trigger a message to the groove object that would initiate the playback. The 'sig~' object was necessary for the groove object to work by transmitting a signal, which is just required to start the playback. We used the X-axis accelerometer sensor on the guitar to control the volume. The sensor sends a value between -1 and 1. When the guitar was raised, the sensor sends a higher negative value that we want to use to increase the volume. The input number was

scaled and then multiplied by -1 to change it to a positive number before being sent to the 'gain~' object. We also used a live gain object to control the audio level of our input signal from our Audio Go Box so Hugh could play over the recording loop. The groove object is not sending the audio playback directly to the 'ezdac~' object, but is instead sending the playback into the Valhalla Supermassive plugin on Max MSP, where the playback audio can be manipulated. The preset values in the Valhalla Supermassive plugin can be manipulated to determine how the audio is augmented. With the desired mix of audio effects, the playback of the recording would sound considerably different to the original signal. To determine when and when not to use the augmented sound effects, Hugh decided to use the accelerometer sensor's Z-axis reading. When the guitar was tilted forward, he wanted the sound effects of the playback to be enabled. Likewise, when the guitar was in its normal position, the playback sound was to be unaffected. Hugh determined through trial and error that a sensor reading of .1 was desirable in determining when to turn on the effect. To control the Valhalla Supermassive plugin, we used the 'Mix' attribute to determine the level of augmentation.

5. REFLECTION ON SYSTEM BUILT

On reflection, the use of the potentiometer could have been an interesting addition to include for manipulating the level of effects that the recording playback was subjected to. In the current system we ended up using, we pre-set the contents of the mix in Valhalla Supermassive to be either fully *on* or fully *off*. An introduction of the potentiometer to this project could have then therefore allowed us to more dynamically adjust the level of the 'Mix' in the performance. Given the audio input being a rhythm guitar section, it was more difficult than initially thought to create the sound effects that we had envisioned for the riff using Valhalla Supermassive as well as other Max MSP plugins we tried. With a deeper understanding of audio signals and how they can be manipulated, we could have further developed our desired effects. For example, we initially wanted to create a 'reverb' effect to the audio playback, which we found to be considerably more difficult using the Valhalla Supermassive and its audio processing settings. Tommy was able to create a patch that outputs some video effects that never made it to the final iteration. The initial idea was to have these effects work in sync with the accelerometer's movement behaviour. He explored some objects that were able to take two input videos / images and "fade" them over one another in response to the accelerometer readings. However, while interesting, we discovered that manipulating a single image or video in a pre-imagined way was more difficult than anticipated, and gave us a realization that there was far more knowledge to be learned on this subject in and of itself. We appreciate that we were able to directly integrate a musical instrument through objects readily available to us in Max MSP, and control the entire performance through the patch, as opposed to needing physical means of manipulating the audio effects. So, our successful exploration of Max MSP's objects ultimately gave us a better appreciation of what the software can do and how it brought our ideas to fruition.

6. PLAN FOR PERFORMANCE

The initial plan for the performance was for Hugh to play the guitar with Tommy sitting next to him to ensure that no issues arose in Max MSP through the computer. Tommy would also be responsible for adjusting the volume of Hugh's guitar input that

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would play back *over* the recorded loop. There was going to be visuals initially, but they were scrapped as we felt it would take away from the overall performance and deviate from our overall intent. Hugh had a general outline for a piece written, but with a lot of room to improvise with the notes and the tools we had created. So, we worked out a choreography of sorts in practice for adjustment of the loop, and Hugh practiced playing while doing these movements that would affect the sensors attached to the bottom front of the guitar. The plan was therefore for Hugh perform an improvised set using the mechanics of the movements learned, and returning to a set riff to bring me back if the improve was not working for a certain section. Initially, Tommy was going to take the jack out of the guitar and plug it into Hugh's amplifier after the loop was recorded, but it was pointed out to us that it would be much more efficient to send the live playing into Max MSP. This set up a small delay in the live playing, but it was ultimately overcome and therefore not noticeable. Furthermore, this change of plan created a much more cohesive performance – as well as project - overall.

7. REFLECTION ON PERFORMANCE PLAN

There were a few factors that did not go to the original plan in the performance. The choreography that we had planned for this performance had made it quite difficult to move and play the improvisation at the same time. The riff that Hugh had practiced on a consistent basis was not affected by the movement of the guitar, but he did find it quite difficult to find the notes and create some interesting sounds while also concentrating on the movements of the guitar. We feel that if we were to do this project again, we would do three things. First, we would focus less on improvisation, and make the body movements more natural as well as make the jump from no mix on the Valhalla Supermassive Plugin to 100 percent mix more gradual, since we felt that this was very harsh. Secondly, the improvisation was a hindrance in the live show, as it was too difficult to do that and focus on movement at the same time. Thirdly, the guitar movements and their functions could have been more intuitive. However, there were also positive factors that went well. We felt

that our performance was still an effective demonstration of how Max MSP can be applied to augmenting instruments for a first venture, and it also showed enough of what we were trying to do that it could be chalked down as a success.

8. FUTURE DEVELOPMENTS

Regarding the takeaways of the content learned in both this course overall and this project - in Tommy's case as an interaction designer - MaxMSP can be particularly useful for designing interactive experiences in both public places or even on websites, where users' interactions with a certain form of content can trigger dynamic responses. Furthermore, Max MSP's large library of objects can also help Tommy as an interaction designer to explore different methods of user action beyond traditional interfaces. More particularly, Tommy takes interest in ideas like movement detection (as also explored in this project) as a concept of interactivity that could further be applied in his future career.

In terms of future development for Hugh, he felt that this project had greatly improved my understanding of Max MSP's capabilities in music, and also his understanding of the things that he takes for granted in an instrument, like something as simple as comfortability using an instrument. In terms of future development for the project itself, we feel that if someone had enough time, as well as resources to invest into this project, that it would be a very useful tool for musicians worldwide to apply. For example, a new instrument that uses natural body movement to alter a track, while also retaining the same entry level skill as a guitar could be very promising in the world of live performance, especially concerning Hugh's interests as a musician.

9. REFERENCES

Benito Temprano, A.L. and McPherson, A. (2021). A TMR Angle Sensor for Gesture Acquisition and Disambiguation on the Electric Guitar. Audio Mostly 2021. [online] doi: <https://doi.org/10.1145/3478384.3478427>.